Day 11 Task — Harden a Linux VM

Time needed	Deadline	Where to submit
60 minutes	Before the Day 12 session	Portal → Day 11 → Upload

Take a VirtualBox snapshot of your Ubuntu VM BEFORE you start (Machine → Take Snapshot). If anything breaks, roll back. Do this only on your own lab VM — never on a machine you don't own.

You'll score your VM with Lynis, harden it, then score it again and prove the improvement. Record the exact commands and the two scores.

Step 1 — Baseline scan with Lynis

Install and run Lynis, then read the summary at the end of the output.

```
sudo apt update && sudo apt install -y lynis
sudo lynis audit system
```

Question	Your answer
Hardening index (the score, e.g. 61) — BEFORE	
How many warnings did Lynis report?	
Write down any THREE suggestions it made	

Step 2 — Apply the hardening steps

Do each item, record the command, and tick it. Read each command before running it.

#	Hardening action	Command (reference)	Done
1	Update all packages	sudo apt update && sudo apt upgrade -y	
2	Enable the host firewall, allow only SSH	sudo apt install -y ufw ; sudo ufw allow 22 ; sudo ufw enable	
3	Install & enable automatic security updates	sudo apt install -y unattended-upgrades	
4	Harden SSH: no root login	edit /etc/ssh/sshd_config -> PermitRootLogin no	
5	Harden SSH: no password auth (use keys)	PasswordAuthentication no ; sudo systemctl restart ssh	
6	Install fail2ban to block brute-force	sudo apt install -y fail2ban	
7	List & disable one unused service	systemctl list-units --type=service ; sudo systemctl disable	

#	Hardening action	Command (reference)	Done
8	Check for empty-password or risky accounts	sudo awk -F: '($2=="") {print}' /etc/shadow	
9	Set a login warning banner	echo 'Authorized access only' | sudo tee /etc/issue.net	
10	Remove one package you don't need	sudo apt remove (e.g. telnet if present)	

Step 3 — Re-scan and compare

Run Lynis again and read the new score.

```
sudo lynis audit system
```

Measure	Value
Hardening index — AFTER	
Points gained (after minus before)	
Did the warning count go down? By how much?	

Step 4 — Two questions (2–3 lines each)

1. Pick ONE change you made and explain, in attacker terms, exactly what it stops.
2. Lynis will never reach 100. Why is 'perfectly hardened' neither possible nor always desirable? (Think about usability and the services the machine must run.)

How to submit

- One Google Doc or PDF with both Lynis scores (screenshots of the summary are ideal), the Step 2 table, and both answers
- Screenshots of the 'Hardening index' line before and after are the key evidence
- Upload under Day 11 on the portal before the Day 12 session

Marks (out of 10)

Part	Marks
Step 1 — baseline score recorded with suggestions	2
Step 2 — hardening steps done with commands	5
Step 3 — after-score and improvement shown	2
Step 4 — reasoning answers	1

Counts toward your lab journal (30%). Marks are for the improvement and your evidence, not for hitting a specific number. Tomorrow we scan this same VM for vulnerabilities with Nmap.